

TEHSIL-LEVEL MICRO CANAL PROJECT

Irrigation Purpose • Command Area Development • Water Distribution

Project focus

- Develop a planned micro-canal network to convey irrigation water from a dependable source to agricultural fields.
- Use smaller distribution channels to improve reach, reduce uncontrolled field flooding and support equitable water delivery.
- Plan the system at tehsil level by grouping villages, command areas, source points and distribution requirements.
- Integrate survey, alignment, hydraulic planning, structures, construction, O&M and monitoring.

Project objective: reliable irrigation water delivery to the identified command area.



Broad micro-canal irrigation command area





1. Need and Objectives of the Micro Canal Project

Why a planned irrigation distribution network is required

Key objectives

- Bring irrigation water closer to agricultural fields and reduce conveyance distance at the field level.
- Improve water distribution among different outlets and tail-end areas.
- Support timely irrigation according to crop and seasonal water requirements.
- Reduce water losses caused by uncontrolled flow, poor alignment and damaged channels.
- Create a maintainable network with defined outlets, inspection points and responsibilities.
- Improve agricultural productivity and resilience where assured irrigation is currently limited.

Planning focus: source → main conveyance → micro canal → outlet → field.



Canal alignment and field distribution



2. Pre-Project Survey and Command Area Assessment

Engineering starts with a reliable field survey

Survey requirements

- Topographic survey of source, proposed canal alignment and command area.
- Ground levels and longitudinal slope along the proposed alignment.
- Existing natural drains, roads, crossings, field boundaries and settlement constraints.
- Soil and foundation conditions for canal bed, side slopes and structures.
- Irrigable command area: land use, crop pattern and irrigation demand.
- Existing irrigation sources and their seasonal reliability.
- Identify head, middle and tail reaches for equitable distribution planning.

Survey output: base map + longitudinal profile + command-area map + preliminary alignment.



Intake-to-field irrigation concept



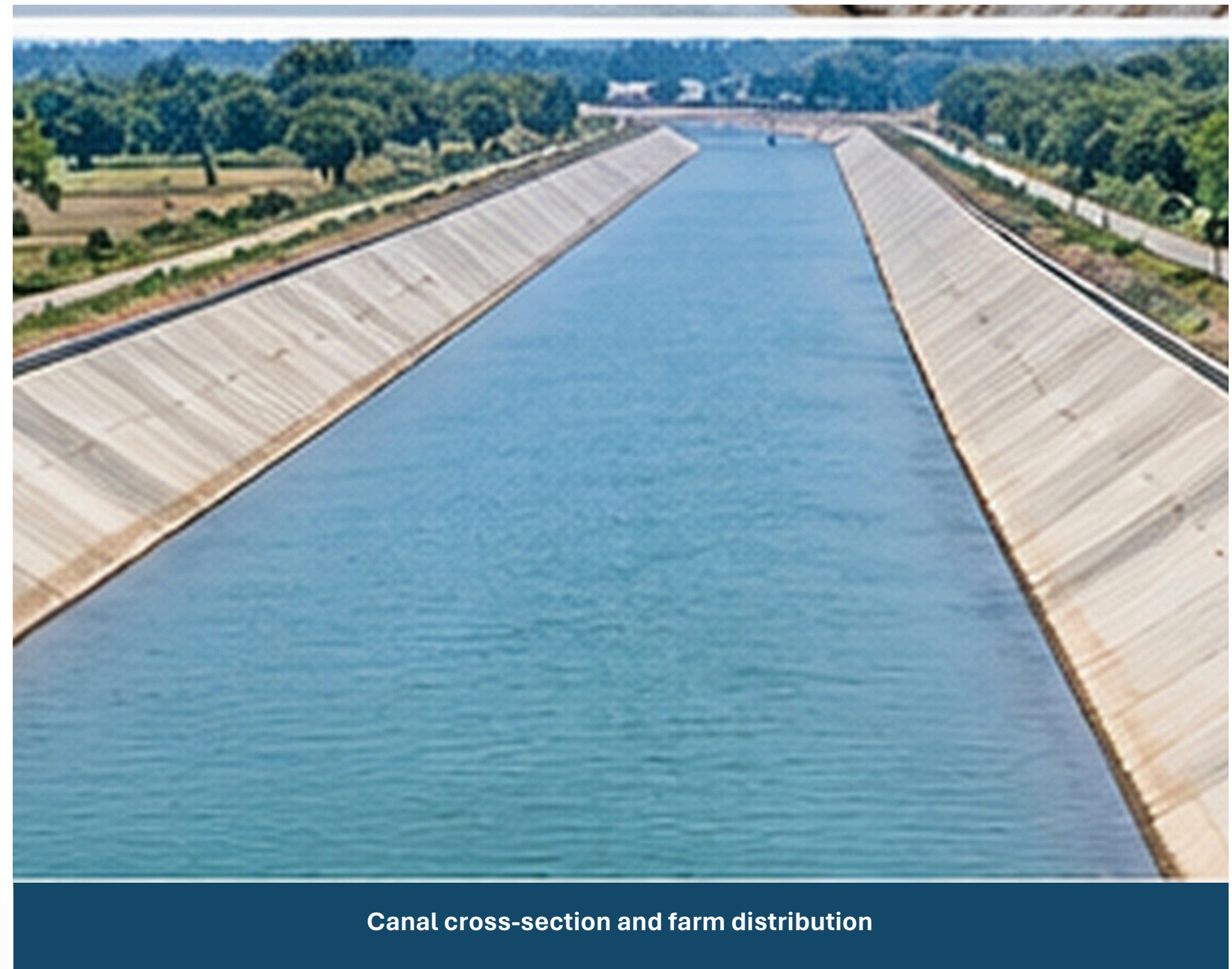
3. Micro Canal Alignment and Design Principles

Follow the natural terrain while protecting the command area

Alignment considerations

- Follow suitable contours and natural ground levels to maintain economical flow.
- Avoid unnecessary sharp bends, unstable ground and locations with repeated erosion risk.
- Provide adequate right-of-way and consider existing roads, drains, utilities and field access.
- Maintain practical access for inspection, desilting and repair.
- Locate branches and outlets according to the actual field distribution pattern.
- Check the complete alignment from head to tail so that downstream fields are not deprived.

Design principle: shortest practical and hydraulically workable alignment—not simply the shortest line.





4. Canal Section and Hydraulic Planning

Size the channel according to planned irrigation demand

Important design parameters

- Design discharge based on irrigation demand, command area, crop pattern and irrigation scheduling.
- Canal bed width, water depth and side slopes must provide stable conveyance.
- Maintain sufficient freeboard above normal operating water level.
- Check velocity so that excessive erosion and sediment deposition are avoided.
- Provide lining where seepage, soil condition or maintenance requirements justify it.
- Design outlets and branches so that water can be distributed without excessive field flooding.
- Final dimensions must be established through the project's hydraulic calculations and approved design.

Final canal dimensions should come from detailed hydraulic design and field levels.



Outlet and branch distribution



5. Irrigation Structures and Water-Control Points

Structures make the micro-canal network controllable

Typical structures to consider

- Intake / head control arrangement at the water source.
- Division and control structures at important branch points.
- Field outlets for controlled delivery to individual command areas.
- Cross-drainage / road-crossing structures where the alignment intersects existing drains or roads.
- Escape or overflow arrangements where required by the hydraulic design.
- Measuring or regulating arrangements where water allocation needs to be monitored.

Every important branch should have a clear control and maintenance arrangement.



Command-area irrigation network



6. Command Area and Irrigation Distribution

Plan the network around farms—not only around the canal

Distribution planning

- Map irrigable fields and identify head, middle and tail reaches.
- Locate outlets close to the areas they are intended to serve.
- Divide the command area into practical irrigation blocks.
- Plan rotational or scheduled distribution where available supply is limited.
- Give attention to tail-end farmers to reduce inequitable water distribution.
- Provide field access and maintenance paths wherever required.

Command-area map + outlet schedule = basis for practical water distribution.



Construction and embankment concept

7. Construction and Quality Control



Build the canal to the approved alignment, levels and section

Construction sequence

- Setting out and centre-line marking from the approved survey.
- Earthwork, excavation and embankment formation.
- Bed preparation and compaction to the specified level and slope.
- Lining or protection works where included in the design.
- Construction of outlets, crossings, control structures and protection works.
- Check levels, dimensions, compaction, concrete/masonry quality and finishing.
- Complete trial flow and rectify leakage, erosion or obstruction before commissioning.



O&M and water distribution

Quality control must be continuous—from setting out to trial operation.



8. Operation, Maintenance and Water Management

Long-term performance depends on regular maintenance

O&M requirements

- Regular desilting and removal of vegetation, debris and obstructions.
- Repair breaches, erosion, damaged lining and outlet defects promptly.
- Keep cross-drainage and road-crossing structures clear.
- Inspect the canal before and during the irrigation season.
- Maintain an outlet-wise water distribution schedule and record major complaints.
- Monitor head-to-tail water availability and correct operational imbalance.
- Assign clear responsibility for routine and emergency maintenance.

O&M target: maintain design flow, prevent damage and ensure dependable tail-end delivery.



Monitoring of irrigated command area



9. Monitoring, Outcomes and Tehsil-Level Management

Measure whether the project is delivering irrigation benefits

Key monitoring indicators

- Length of micro canal completed and commissioned.
- Command area brought under reliable irrigation.
- Number and functioning status of outlets and control structures.
- Water availability at head, middle and tail reaches.
- Area irrigated during the season and crop coverage.
- Canal losses, recurring breaches and maintenance requirements.
- Farmer feedback, equitable distribution and O&M performance.
- Annual review of water demand, crop pattern and system capacity.

Expected outcome: reliable irrigation + better distribution + improved command-area utilisation.

